

ASSESSMENT-3

Unit - 1

Name: JOSEPH JESWIN.C

Reg no: 192511094

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Automation System Using Intelligent Agents.

Introduction

A Smart Home Automation System uses Artificial Intelligence to automatically control home appliances. ~~Such~~ such as lights, air conditioners, fans and security. It uses sensors like motion detector, temperature sensors and cameras to monitor the environment and make intelligent decisions.

Types of Intelligent Agents

1. Simple Reflex Agent

A Simple Reflex Agent makes decision based only on the current input using condition-action rule.

Example:

Motion detected $\rightarrow$ Turn on lights

No Motion $\rightarrow$ Turn off

Smoke detected $\rightarrow$ Activate fire alarm.

2. Model - Based Reflex Agent

A Model - Based Reflex Agent stores information about the environment and remember previous states.

Example :

- Remembers whether light are already on
- Knows room occupancy
- Tracks Indoor temperature.

3. Goal - Based Agent.

A Goal Based Agent chooses action to achieve a specific goal.

Goals in Smart Home.

- Maintain comfortable temperature
- Reduce electricity consumption
- Ensure home security.

Example

If the room become hot, it decides whether to start the AC or not to reach the desired temperature.

4. Utility - Based Agent

A Utility - Based Agent selects the action that provides the maximum overall benefit.

considers;

Comfort

- Energy saving
- Cost
- Security.

Examples

If opening the window looks the room naturally, it chooses that instead of using the Ac to save electricity.

Comparison

	Memory	Goal	Best use
Simple Reflex	No	No	Immediate use
Model - Based	Yes	No	Monitoring environment
Goal - Based	Yes	Yes	Planning actions
Utility - Based	Yes	Yes	Optimizing Comfort

Conclusion

A Hybrid Intelligent Agent is the most effective choice because it responds quickly, remembers previous situation plan intelligently, and learns user preferences. This ensures maximum comfort, better security and reduced energy consumption.

2. Robot Navigation in an Unknown Maze and IDs.

Introduction

A Robot placed in an unknown Maze must find the exit without knowing the layout. Since no prior information is available, Uninformed Search Algorithms such as Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) are used to explore the maze efficiently.

Uniform Cost Search

Definition

Uniform Cost Search expands the node with lowest cumulative path cost first.

Working

- Starts from the initial position
- Calculates the total of each possible path
- Continues until the exit is found.

Advantages

- Complete algorithm
- Finds the least-cost path

Advantages

Low memory usage

Slower because many nodes are explored

Iterative Deepening Search

Definition

(IDS) Repeatedly performs Depth-limited search with increasing depth limits until the goal is found.

Working

- Search at depth 0
- If goal not found, search at depth 1
- Continue increasing the depth until the exit is reached.

Advantages

- Requires very little memory
- Complete for finite search spaces.

Disadvantages

- Repeats node expansion
- Less efficient than UCS when path costs differ.

Complexity Comparison

Feature	UCS	IDS
Complete	Yes	Yes
Optimal	Yes	only when costs are equal
Time Complexity	High	$O(b^d)$
Space Complexity	High	$O(b \times d)$
Memory Usage	High	low

Conclusion

UCS is ideal when path costs vary, while IDS is better when memory is limited. For most real-world robot navigation problems, a goal-based agent with UCS is the most effective solution.